Cover letter

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Unsent submission correspondence; prepared for author review.

Dear Editors,

Please consider the manuscript "The sl(2) weight system at c = -3/32 extends to a graph 4-invariant" by Deep Bhattacharjee and Ushashi Bhattacharya.

The paper proves Conjecture 4.1 of Fomichev, Karev, Pavutnitskiy and Usanov: the sl(2) weight system at c=-3/32 extends to a graph four-invariant. The explicit signed-count construction works on every finite graph and agrees with the matrix-defined weight system on every chord diagram. This revision adds a normal-form proof and clarifies verification scope.

The explicit graph invariant and its links with Lie algebra weight systems fit the journal's combinatorial and algebraic scope. The neighbouring c=3/8 specialization was treated in this journal.

The prior public preprint is https://hexagonmath.org/2610.00320v1. The research manuscript/source archive is https://doi.org/10.5281/zenodo.23541142. Editable sources and verification records are supplied with the manuscript.

The universal extension theorem and its matrix-trace definition are formalized in Lean 4.34.1 with pinned Mathlib. At the unchanged proof-source commit, the complete build and four endpoint axiom reports passed with only propext, Classical.choice and Quot.sound. Python exact checks were rerun locally; Julia, C and C++ passed in the archived full run. The grouping proof for part (a) and the added faithful-action lemma are supplied in the written argument; no new formalization of the abstract Weyl algebra is claimed.

The manuscript and the accompanying declaration disclose the substantive use of Claude and OpenAI tools. The existing public provenance records mixed human and AI text and partial human checking; the author reviews any updated declaration at submission. The named human authors retain responsibility for the content.

Sincerely,

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